

Watcher v2 — Build Roadmap

From a message filer to a receptionist. Every item scored for urgency and ease. - 14 August 2026 - v1.13

The whole gap in one line: the pipeline can listen and file, but it cannot reply. Three outcomes exist and all three mean "put this somewhere". Adding a fourth — **answer the customer** — is the project.

Where we stand today

Built and tested	Missing
248 passing tests, no DB or network needed	Knowledge — zero tables, zero rows
98 Python files, 13 modules	Live availability — no read path to any PMS
18 DB tables + 3 migrations	Proof of identity — matching is not verifying
Every external system behind a swappable seam	Intent taxonomy unification — classification and vocabulary use different enums
Eval runner + job queue, with a CI accuracy gate	Prompt v2 — golden set still at 8 examples
Receptionist vocabulary: 19 intents, data not code	
Channel-neutral envelope + WhatsApp/voice adapters	
Task state machine and the autonomy gate	
Reply path wired — RECEPTIONIST_REPLY is the fourth outcome	
Persistence — conversations, turns, tasks in Postgres	
Channel-neutral core — KNOWN_LEAKS = {}, boundary test enforces it	
Python 3.13	

How to read the scores

- **NOW** — Today or tomorrow. Something is actively bleeding, or it blocks other work.
- **HIGH** — This sprint. The demo does not exist without it.
- **MED** — Needed before a real client, not before a demo.
- **LOW** — Do it when convenient. Nothing waits on it.
- **Trivial** — Minutes. A setting, a click, a delete.
- **Easy** — Understood problem, no design decisions left open.
- **Moderate** — Real work, but the pattern already exists in the repo to copy.
- **Hard** — Genuinely new. Nothing in the repo to copy from.

Totals: ~6.5 engineering days remaining by the numbers below. Track 0 is done, Track 1 is done, and the core of Track 2 (2.1, 2.2, 2.3) is done. At six days a week that is a demo in about one week and a real client's guests on it in about two. The critical path is now **2.4 (knowledge base)** and **3.1 (PMS read port)**: no facts means no answers, and no read path means no pricing, no availability and no door codes.

Track 0 — Today. Hours, not days. [COMPLETE]

All five items done.

#	Work item	Urgency	Ease	Days	Status
0.1	Set default branch to main	NOW	Trivial	1 min	DONE
0.2	Remove the client name from the golden set and fixtures	NOW	Easy	0.25	DONE
0.3	Decide the receptionist intent vocabulary	NOW	Easy	1 hr	DONE — 19 intents, 84 examples, 5 languages, 38 tests
0.4	Merge the eval branch	NOW	Trivial	0.25	DONE
0.5	Delete the stale nifty-johnson branch	LOW	Trivial	1 min	DONE

Track 1 — Foundations. Week 1. [COMPLETE]

All three items done.

#	Work item	Urgency	Ease	Days	Status
1.1	Stop the core speaking WhatsApp	NOW	Moderate	1.5	DONE — <code>wa_message_id</code> to <code>external_id</code> , <code>wa_chat_id</code> to <code>thread_id</code> , <code>sender_wa_name</code> to <code>sender_display_name</code> . Added <code>channel</code> field. <code>KNOWN_LEAKS = {}</code> . Alembic migration 002. <code>MetaSettings</code> moved to <code>channels/whatsapp.py</code> . <code>to_inbound_turn()</code> bridge added. <code>CHANNEL_REGISTRY</code> exempts legitimate data values.
1.2	Port the four kept scaffold files	HIGH	Moderate	1.0	DONE (prior session)

1.3	Python 3.12 to 3.13	LOW	Easy	0.5	DONE — 7 version pins updated across pyproject.toml, packages, and CI.
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Track 2 — Make it a receptionist. Week 2. [CORE DONE, 2.4 and 2.5 REMAIN]

The product's reply path is wired end-to-end. What remains is the knowledge base and prompt tuning.

#	Work item	Urgency	Ease	Days	Status
2.1	Conversations, tasks and slot filling	HIGH	Hard	2.0	DONE — 7 new tables via Alembic migration 003: <code>contacts</code> , <code>conversations</code> , <code>turns</code> , <code>task_rows</code> , <code>understandings</code> , <code>corrections</code> , <code>usage_events</code> . ORM models in <code>db/models.py</code> . <code>ConversationRepository</code> with <code>find_or_create_conversation</code> , <code>record_turn</code> , <code>get_active_task</code> , <code>save_task</code> , <code>create_task</code> . <code>task_from_row</code> / <code>task_to_row</code> converters. Verification-codes table deferred to 2.4 (knowledge base).

2.2	The reply path	HIGH	Moderate	1.5	DONE — Tool registry (<code>TakeMessage</code>, <code>HandoffToHuman</code>) validated against vocabulary <code>terminal_tool</code>. Receptionist function in <code>conversations/receptionist.py</code>: check autonomy, manage task state, return <code>OutboundAction</code> (ask/confirm/say/hand off). <code>ChannelSender</code> protocol in <code>channels/sender.py</code>. 8 tests.
2.3	Autonomy gate	HIGH	Easy	1.0	DONE — <code>RECEPTIONIST_REPLY</code> as fourth <code>RoutingAction</code>. <code>Receptionist</code> protocol in worker. After classification + identity resolution, <code>decide_autonomy()</code> runs before rule matching. Acting intents invoke receptionist; hand-off falls through to existing routing. <code>ProcessOutcome</code> carries <code>autonomy</code> and <code>outbound_action</code>. 4 new orchestration tests.
2.4	Knowledge base	HIGH	Moderate	2.0	NEXT — Facts table with sensitivity flags, prose in pgvector, and a real "I don't know" that fetches a human. Door codes are not ordinary facts. For the demo, one property's facts fit in the prompt — no retrieval needed.

2.5	Prompt v2 and rewrite the golden set	MED	Moderate	1.0	PENDING — Unblocked: 0.3 is done. Receptionist vocabulary, 8 cases rewritten and ~50 added. The runner from 0.4 already exists to score them. Watch the mixed-language number — it is 0% on the current 8.
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Known design gap: intent taxonomy unification

`IntentType` (the classification enum: `new_lead`, `support_issue`, etc.) and the vocabulary intents (`property_question`, `booking_enquiry`, etc.) are separate taxonomies. The autonomy gate returns `hand_off` for any intent not in the vocabulary, so the receptionist path only fires when the classifier produces a vocabulary-recognized intent. **Unifying the two is prerequisite to the receptionist handling real traffic.** Estimate: 0.5 days, Easy — the vocabulary already declares the canonical set; the classification enum needs to adopt it.

#	Work item	Urgency	Ease	Days	Status
2.6	Intent taxonomy unification	NOW	Easy	0.5	NEW — Merge <code>IntentType</code> enum with vocabulary intent names so <code>decide_autonomy()</code> recognises classified intents and the receptionist fires on real messages.

Track 3 — Integration and launch. Week 3.

Base360.ai is ruled out as a partner: their product is substantially ours and their client base is closed to us. Hostaway, Guesty and Cloudbeds all publish APIs, so the capability is available without the strategic cost.

#	Work item	Urgency	Ease	Days	Status
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3.1	PropertySystemPort plus the first adapter — the bottleneck	MED	Moderate	2.5	PENDING — Build the port, not a Hostaway integration. Blocks pricing, availability and door codes: with no way to look up a booking, the identity check cannot run. crm_cache is the right shape; delivery is write-only, so this needs a new read port. Cache facts; never cache availability.
3.2	End to end on a real number, then measure	HIGH	Moderate	1.0	PENDING — The point at which the eval number becomes real rather than recorded.

Runs in parallel — start on day one

None of these are engineering. All of them can quietly become the reason a date slips.

#	Work item	Urgency	Ease	Days	Status
P1	Pick the first client	NOW	Easy	--	Decides which PMS adapter gets built first. Without it, 3.1 is a guess.
P2	Read the Hostaway / Guesty / Cloudbeds API docs, get sandbox keys	HIGH	Easy	0.5	Public docs, usually self-service sandboxes. Shape the port from what two or three of them offer in common.
P3	File Meta business verification	MED	Easy	1 hr	No longer blocking — unverified start is allowed, and a receptionist mostly replies inside the 24-hour window. File it anyway before volume makes it bind.

P4	Graphify as a build aid (optional)	LOW	Easy	0.5	Maps the repo for the coding agents during the 1.1 rename. Local parsing, no vector store, nothing leaves the machine.
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The dates (revised 14 August 2026)

Milestone	Original	Revised	Status
Eval merged, name leak fixed, vocabulary decided	End of day 1	--	DONE
Core stops speaking WhatsApp; scaffold files ported	End of week 1	--	DONE
Holds a conversation and replies — rough demo	Middle of week 2	End of this week	DONE (wiring complete, needs taxonomy unification + knowledge for a real demo)
Knowledge and safety rules in — demo-ready	End of week 2	~2 days from now	Blocked on 2.4 + 2.6
Live availability, measured, on a real number — pilot-ready	Middle of week 3	~5-6 days from now	Blocked on 3.1 + P1
Phone answering	Week 4	Week 4	Unchanged — speech-to-text seam already exists

What would actually move these dates

Intent taxonomy unification drifting. Half a day of work that the receptionist literally cannot function without on real traffic. It is the new 0.3 — small, high leverage, do it first.

Knowledge base scope creep. The 2.0 days assumes a structured intake form plus PMS sync. "Can it read our PDF handbook?" is a different and much larger project. Say no for now.

Quality, not features. Getting to *working* is days away. Getting to *trustworthy* — where it never invents a check-in time — is measured by the eval runner, and that number is not fully under your control. Budget a tuning tail.

PMS API access. Docs are public, but sandbox approval and rate limits are theirs to grant. Start P2 on day one; it is the new version of the Meta-verification lesson.

If you do only one thing today: unify the intent taxonomy (2.6) and start the knowledge base (2.4). One is half a day of enum refactoring, the other is the last hard engineering item before a demo. Together they make the receptionist real.

Session log — 14 August 2026

PR: #13 on [claude/file-review-planning-dcyb2g](#) **Tests:** 228 to 248 (+20 new tests) **Commits:** 5 (one per item: 1.1, 1.3, 2.1, 2.2, 2.3) + docs

Items completed this session: 1.1 (1.5d), 1.3 (0.5d), 2.1 (2.0d), 2.2 (1.5d), 2.3 (1.0d) = **6.5 engineering days delivered.**